

Measuring the catalase activity:

Understanding the enzyme-catalyzed reaction

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Abstract

Enzymes are vital to most biological functions and can be found in most organisms. As catalysts, they can hasten the chemical reaction. In this experiment, we examined the existence of catalase, an enzyme that can catalyze the breakdown of hydrogen peroxide, in six different vegetables and measured the rate of the enzyme-catalyzed reaction by measuring the volume of the oxygen produced by the breakdown within the first minute. We found that most raw fruit or vegetables contained certain substance that can hasten the breakdown of hydrogen peroxide, which could provide evidence for the existence of enzymes in these vegetables. This experiment also indicated that the capacity to speed up the reaction varied according to the type of the vegetables.

Introduction

Enzymes are macromolecules that can speed up the chemical reaction without being consumed. (Campbell et al., 2016) Hydrogen peroxide is a kind of reactive oxygen substance in most organisms exposed to oxygen. (Chelikani, Fita and Loewen, 2003) The reactive oxygen substance which was once regarded toxic to plants, is related to many important procedures such as programmed cell death. The substance is therefore tightly regulated. (Mittler, 2002) Catalase is an enzyme commonly seen in organisms that can catalyze the decomposition of hydrogen peroxide ($2 \text{H}_2\text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{O}_2$). (Chelikani, Fita and Loewen, 2003) Yet the activity of catalase was not familiar to most of us students. Whether this enzyme exists in all organisms and how the enzyme affects the reaction were what we wanted to find out in this experiment.

Aims & Hypothesis

We aimed to measure the rate of breakdown of hydrogen peroxide by measuring the volume of oxygen produced by the reaction within one minute after the reaction began. We hypothesized that catalase in the raw vegetables could hasten the breakdown of hydrogen peroxide. We also hypothesized that how much the enzyme could speed up the reaction varied in different organisms.

Methods

All procedures were carried out according to the protocol that had been assigned to us before the class. During the experiment, we wore gloves to avoid the potential harm to the skin caused by hydrogen peroxide. We were provided by the lab with 6 kinds of fruits or vegetables as our experimental materials. Among them, 5 were raw (apple, banana, garlic, onion and raw cabbage) and one was boiled (boiled cabbage). Hydrogen peroxide was also provided by the lab.

To prepare for the experiment, we weighed out around 10g of each kind of the fruit/vegetables. Then, we add 10g of one kind of the fruit/vegetables, along with 30mL water, into a blender respectively, to cut them up until no chunks can be seen. In the meantime, my partner made up the apparatus (see Figure 1) according to the protocol and the illustration made by the instructors in the lab.

In order to measure the rate of the reaction, we targeted to measure the volume of oxygen produced by the reaction within the first minute. In this experiment, we used a set of apparatus to measure the volume. (see Figure 1) The experimental apparatus was hermetic. A syringe was used to measure and push 20 cm³ hydrogen peroxide into the conical flask, where 5g of blended vegetable was placed. As soon as hydrogen peroxide was pushed into the flask, we started timing for the reaction. The conical flask was linked to a measuring cylinder filled by water which was put reversely in a bowl with half of water inside to ensure the hermetization of the apparatus.

To measure the volume of the oxygen produced by the reaction, we read the volume taken up by the gas in the cylinder, for when hydrogen peroxide broke down, the oxygen produced by the reaction would be conducted to the measuring cylinder. With one person using a timer and one person operating, we were able to know how much oxygen was produced within 1 minute.

For each kind of fruit/vegetables, we repeated the procedures above, including preparation, assembling, making the reaction happen and measurement, in order to get data from different kinds of vegetables. We recorded all the data and uploaded to a computer when we finished all the measurement, so that we could share the data.

Results

We received gas in the measuring cylinder when using apple, banana, garlic and raw cabbage as catalysts respectively, while there was no gas received when using boiled cabbage and onion. Within 1 minute, we received 14mL of gas when using raw cabbage, 3mL when using garlic, 2mL when using apple, 1mL when using banana. We received no gas when using onion and boiled cabbage. (Figure 2-1)

Discussion

In this experiment, our hypotheses proved to be correct. We received no gas when using boiled cabbage as a catalyst. In contrast to the raw cabbage, the data from the boiled cabbage indicated that there was some substance failed to assist the decomposition of hydrogen peroxide after boiled, which provided evidence for the existence of enzymes in the plants because boiling provides a high temperature that can inactivate most enzymes. The boiled cabbage could serve as a control of the experiment, indicating that apple, banana, garlic and raw cabbage could hasten the

decomposition of the hydrogen peroxide while onion could not. Therefore, the reaction rate varies according to the type of the fruit or vegetables.

However, the conclusion also gave rise to a new question. What affected the variation in the reaction rates became doubtful. The concentration of the substrate (H_2O_2) remained the same in the experiment. The amount of the catalase in the plants seemed to be the only variable that could possibly affected the reaction rates. Our hypothesis here proved to be close to the truth because there were studies showing that when the substrate concentration is high enough, the reaction rate increases with enzyme concentration increasing. (Uddin, 2012; Katsumata, 1969)

When comparing our data with class average (See Figure 2-2), most of our results were not radically against the class average levels. However, the volumes of the gas we received when using banana and onion as catalysts was far below the class average. We suspected that the long-time exposure to the air caused some chemical reaction in the vegetables inactivating the catalase, yet this needed further research.

There were also other limitations of the experiment. Firstly, we presumed that the reaction was catalyzed by some enzyme, yet actually we could not prove this through the experiment. All we could know was that some substance in certain plants could speed of the breakdown of hydrogen peroxide and that this substance could be ineffective after boiled. Without further research, we cannot know for sure that the reaction was catalyzed by catalase. Secondly, our group failed to weigh out 1.5 cm^3 of boiled cabbage, because some of it was dropped down to the ground by accident while the lab did not have more boiled cabbage. However, its effect on the result should be limited because the catalase was considered to have lost most biochemistry functions after boiled. Although the class average of the volume when using boiled cabbage was 1.5 mL, yet most groups actually received no gas at all, as is shown in Figure 2-1. Thirdly, we did not repeat the experiment, which could add fluctuation to our data and made the average figures less reliable.

Conclusion

To summarize, we found that most vegetables we used could increase the rate of the breakdown of hydrogen peroxide, providing evidence for the existence of enzymes in these vegetables. Also, different vegetables proved to vary in the capacity to speed up the reaction, proving the concentration of enzymes could affect the reaction rate. This experiment tested many basic concepts related to the enzymes, shedding light for several general directions for research in the related fields such as enzyme kinetic to us and enhancing our understanding of enzyme-catalyzed reactions.

Reference

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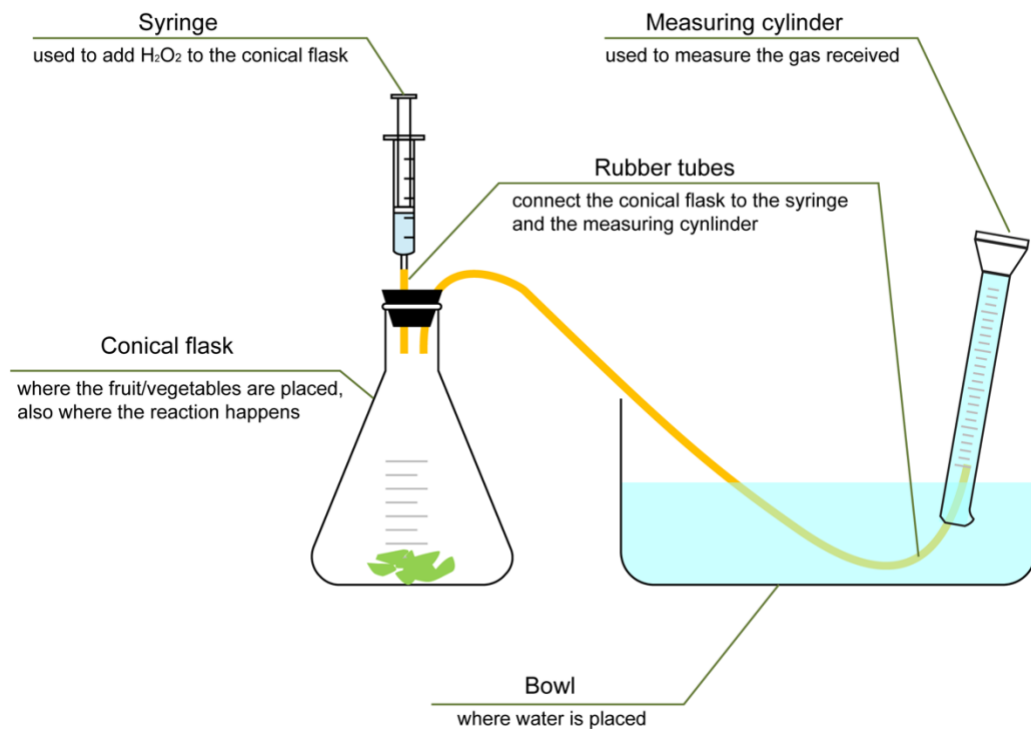


Figure 1. An illustration of the apparatus used in this experiment. (own work created with Adobe Illustrator, according to the descriptions in the protocol)

We used a set of apparatus to measure the volume oxygen we received, which consisted of two major parts. The first part was a conical flask with two holes in its bung; two rubber tubes in the two holes respectively link themselves to a syringe and a measuring cylinder, the latter of which was placed upside down and filled with water in the second part – a bowl around 50% full of water. The tube was connected through the water into the cylinder. We pushed the hydrogen peroxide in the syringe into the conical flask to make the breakdown happen in the flask, catalyzed by the vegetables placed in the bottom of the flask. The oxygen produced in the process will increase the pressure inside to push equal amount of gas into the measuring cylinder through the rubber tube. Then, we could read the volume of gas we received in the cylinder to know how much oxygen was produced.

	<i>Apple</i>	<i>Banana</i>	<i>Garlic</i>	<i>Onion</i>	<i>Cabbage (raw)</i>	<i>Cabbage (boiled)</i>
<i>My group</i>	2	1	3	0	14	0
<i>A</i>	0	0	0	0	18	0
<i>B</i>	2	0	10	1	6	3
<i>C</i>	3	10	3	6	7	0
<i>D</i>	1	9	6	0	N/A	0
<i>E</i>	0	10	0	1	5	0
<i>F</i>	2	12	0	0	26	0
<i>G</i>	1	4	5	1	19	1
<i>H</i>	0	3	0	8	4	0
<i>I</i>	45	94	89	6	3	4
<i>J</i>	1	8	0	2	16	N/A
<i>K</i>	0	5	3	0	4	0
<i>L</i>	0	8	8	2	6	1.5
<i>M</i>	3	15	N/A	2	16	N/A
<i>N</i>	1.5	10	10	15	N/A	N/A
Average	0.9	5.0	2.9	1.5	10.4	0.5

Figure 2-1. Data collected from the experiment N/A means that the data was not available, which could be due to students' failure to carry out the experiment. Each group contains 2-3 individuals. To ensure the anonymity of the report, we use letters (such as A, B, C, D and etc.) to refer to different groups. We did not take Group I in account when calculating the averages, because the data from them was very different from those from all the other groups, which greatly added to the fluctuation of the statistics and was likely to be owing to certain flaws in their method.

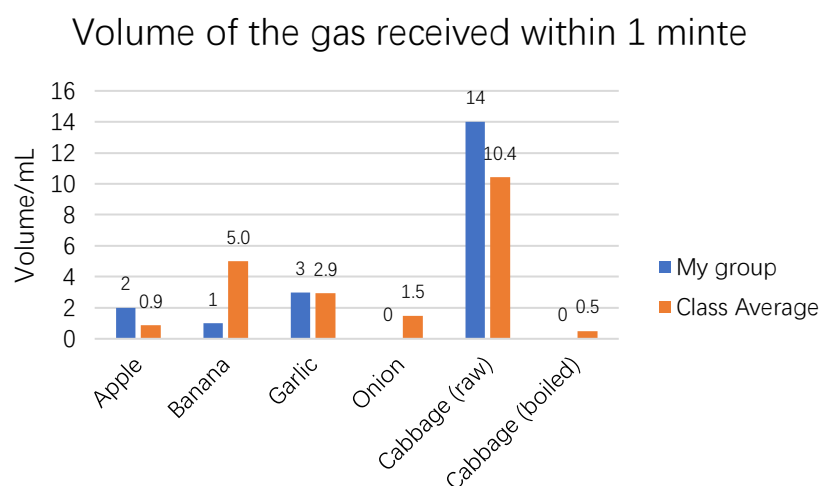


Figure 2-2. The chart based on the data of Figure 2-1 to compare the data from my group with the class average. In this chart, we can see that the volume of gas our group received when using banana and onion was below the class average.